4) public spaces/stairs, 5) engine-rooms and auxiliaries' room, 6) garages and 7) cargo spaces. All have to be divided by main vertical zone.

The calculation criteria for complex groups (e.g. cabin lighting and power sockets) subgroups are to be included via an explanatory note, indicating the load composition (e.g. lights of typical cabins, TV, hair dryer, fridge, etc., typical cabins).

4.1.10 L Entertainment services
This group includes all Loads related to the enterta inment services that typically are: public spaces audio and video equipments, theatre stage equipments, IT systems for offices, video games, etc.

4.1.11 N Cargo Loads

---CHUNK---

SOURCE: MEPC.212(63).pdf
This group includes all Loads related to the enterta inment services that typically are: public spaces audio and video equipments, theatre stage equipments, IT systems for offices, video games, etc.

4.1.11 N Cargo Loads
This group will contain all cargo loads such as cargo pumps, cargo gear, maintaining cargo, cargo reefers loads, cargo hold fans and garage fans for sake of transparency. However, the service factor of this group is to be set to zero.

4.1.12 M Miscellaneous
This group will contain all loads which have not been associated to the above-mentioned groups but still are contributing to the overall load calculation of the normal maximum sea load.

Loads description

4.2 This identifies the loads (for example "seawater pump").

Loads identification tag

4.3 This tag identifies the loads according to the shipyard's standards tagging system. For example, the "PTI1 fresh water pump" identification tag is "SYYIA/C" for an example ship and shipyard. This data provides a unique identifier for each load.

Loads electric circuit Identification

4.4 This is the tag of the electric circuit supplying the load. Such information allows the

---CHUNK---

SOURCE: MEPC.212(63).pdf
and shipyard. This data provides a unique identifier for each load.

Loads electric circuit Identification

4.4 This is the tag of the electric circuit supplying the load. Such information allows the data validation process.

Loads mechanical rated power "Pm"

4.5 This data is to be indicated in the document only when th electric load is made by an electric motor driving a mechanical load (for example a fan, a pump, etc.). This is the rated power of the mechanical device driven by an electric motor.

Loads electric motor rated output power [kW]

4.6 The output power of the electric motor as per maker's name plate or technical specification. This data does not take part of the calculation but is useful to highlight potential over rating of the combination motor-mechanical load.

Loads electric motor efficiency "e" [/]

4.7 This data is to be entered in the document only when the electric load is made by an

---CHUNK---

SOURCE: MEPC.212(63).pdf
potential over rating of the combination motor-mechanical load.

Loads electric motor efficiency "e" [/]

4.7 This data is to be entered in the document only when the electric load is made by an electric motor driving a mechanical load.

Loads rated electric power "Pr" [kW]

4.8 Typically the maximum electric power abs orbed at the load electric terminals at which the load has been designed for its service, as indicated on the maker's name plate and/or maker's technical specification. When t he electric load is made by an electric motor driving a mechanical load the load's rated electric power is: Pr=Pm/e [kW].

Service factor of load "kl" [/]

4.9 Provides the reduction from the loads rated electric power to loads necessary electric power that is to be made when the load absorb less power than its rated power. For example, in case of electric motor driving a mechanical load, a fan could be designed with some power margin, leading to the fact that the fan rated mechanical power exceeds the power requested by the duct system it serves. Another example is when a pump rated

---CHUNK---

SOURCE: MEPC.212(63).pdf
with some power margin, leading to the fact that the fan rated mechanical power exceeds the

power requested by the duct system it serves. Another example is when a pump rated power exceed the power needed for pumping in its de livery fluid circuit. Another example in case of electric self-regulating semi-conductors electric heating system is oversized and the rated power exceeds the power absorbed, according a factor kl.

Service factor of duty "kd" [/]

4.10 Factor of duty is to be used when a function is provided by more than one load. As all loads have to be included in the EPT for EEDI, this factor provides a correct summation of the loads. For example when two pumps serve the same circuit and they run in duty/stand-by their Kd factor will be ½ and ½. When three compressors serves the same circuit and one runs in duty and two in stand-by, then kd is 1/3, 1/3 and 1/3.

Service factor of time "kt" [/]

4.11 A factor of time based on the shipyard's evaluation about the load duty along 24 hours of ship's navigation as defined at paragraph 3. For example the Entertainment loads operate

---CHUNK---

SOURCE: MEPC.212(63).pdf
Service factor of time "kt" [/]

4.11 A factor of time based on the shipyard's evaluation about the load duty along 24 hours of ship's navigation as defined at paragraph 3. For example the Entertainment loads operate at their power for a limited period of time, 4 hours out 24 hours; as a consequence kt=4/24. For example, the seawater cooling pumps operate at their power all the time during the navigation at Vref. As a consequence kt=1.

Service total factor of use "ku" [/]

4.12 The total factor of use that takes into consideration all the service factors: ku=kl·kd·kt.

Loads necessary power "Pload" [kW]

4.13 The individual user contribution to the auxiliary load power is Pload=Pr·ku.

Notes

4.14 A note, as free text, could be included in the document to provide explanations to the verifier.

Groups necessary power [kW]

4.15 The summation of the "Loads necessary power" from group A to N. This is an

---CHUNK---

SOURCE: MEPC.212(63).pdf
Notes

4.14 A note, as free text, could be included in the document to provide explanations to the verifier.

Groups necessary power [kW]

4.15 The summation of the "Loads necessary power" from group A to N. This is an intermediate step which is not strictly necessary for the calculation of PAE. However, it is useful to allow a quantitative analysis of the PAE, providing a standard breakdown for analysis and potential improvements of energy saving.

Auxiliaries load's power PAE [kW]

4.16 Auxiliaries load's power PAE is the summation of the "Load's necessary power" of all the loads divided by the average efficiency of the generator(s) weighted by power.

PAE=Pload(i)/(average efficiency of the generator(s) weighted by power)